

Quiz # 02 (2)**Course Title:** Microprocessors and Assembly Language
Course ID: CSE - 4503**Total Marks:** 15**Time:** 25 Mins.**Student Id:****Obtained Marks:**

- Q1. What is Implied Addressing? Explain the 20-bit address calculation process with appropriate examples for Direct and Indirect Addressing (both for 8-bit and 16-bit data addressing mode). **03**
- Q2. Derive the machine code for the instruction: **IN AL, F1 h**. Also, show how the derived contents can be stored in different memory locations. **03**
- Q3. Your given Student Id format is **22 . .XXXX**. Use last 4 (four) digits of your id as a hexa-decimal value (i.e., **XXXXh**) and consider the following table. **09**

RM \ MOD	MOD			
	00	01	10	11
				W = 0 W = 1
000	[BX] + [SI]	[BX] + [SI] + d8	[BX] + [SI] + d16	AL AX
001	[BX] + [DI]	[BX] + [DI] + d8	[BX] + [DI] + d16	CL CX
010	[BP] + [SI]	[BP] + [SI] + d8	[BP] + [SI] + d16	DL DX
011	[BP] + [DI]	[BP] + [DI] + d8	[BP] + [DI] + d16	BL BX
100	[SI]	[SI] + d8	[SI] + d16	AH SP
101	[DI]	[DI] + d8	[DI] + d16	CH BP
110	d16 (direct address)	[BP] + d8	[BP] + d16	DH SI
111	[BX]	[BX] + d8	[BX] + d16	BH DI

Now, using the table derive the machine code contents of the following MOV instructions and show how many memory locations are required to store the derived machine codes:

- MOV AL, [XXXXh]
- MOV SS: [BX+DI+ XXXXh], DX
- MOV BX, XXXXh